

GB656: Team Assignment 1

Google Colab Notebook:

https://colab.research.google.com/drive/15QLO8lpJJTJum81PpS2ajBotq1_sncMR?usp=sharing

Dataset:

The dataset consists of accounting data from 8,262 companies listed on the New York Stock Exchange (NYSE) and NASDAQ, spanning from 1999 to 2018. The goal of this assignment is to predict the likelihood of a company's bankruptcy using financial ratios, including the Altman Z-score model, and to assess if better predictors can be identified. The dataset comprises 78,682 observations of firm-year combinations.

Altman Z-Scores:

The Altman Z-score formula, first published in 1968, is a widely used model for predicting corporate bankruptcy. It relies on five financial ratios:

Ratios:

- i. **X1:** Working capital / Total assets
- ii. **X2:** Retained earnings / Total assets
- iii. **X3:** EBIT / Total assets
- iv. **X4:** Market value of equity / Total liabilities
- v. **X5:** Sales / Total assets

The Z-score is calculated as follows:

$$Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5$$

This model classifies companies into three zones:

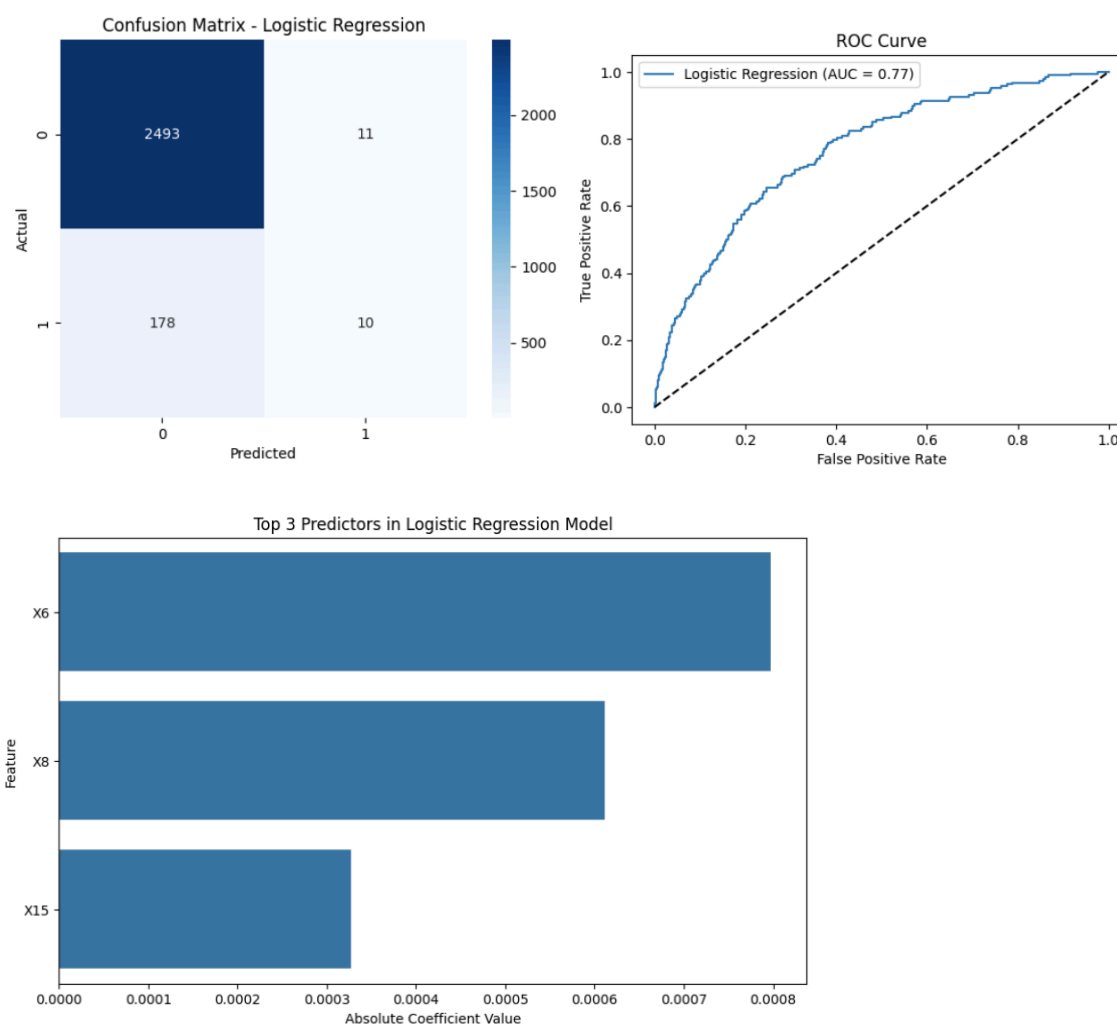
1. **$Z > 2.99$:** Safe zone (low risk of bankruptcy)
2. **$1.81 < Z < 2.99$:** Grey zone (moderate risk)
3. **$Z < 1.81$:** Distress zone (high risk of bankruptcy)

Performance of Altman Z-Score on Dataset

Using the provided dataset, we implemented the Altman Z-score model to predict bankruptcy events in companies. The Z-score demonstrated a moderate ability to classify bankrupt and non-bankrupt companies, with an accuracy of approximately 66% for the test set (2015-2018). However, its performance was influenced by market conditions, particularly during financial crises, where predictions were less reliable due to fluctuations in market value (X4).

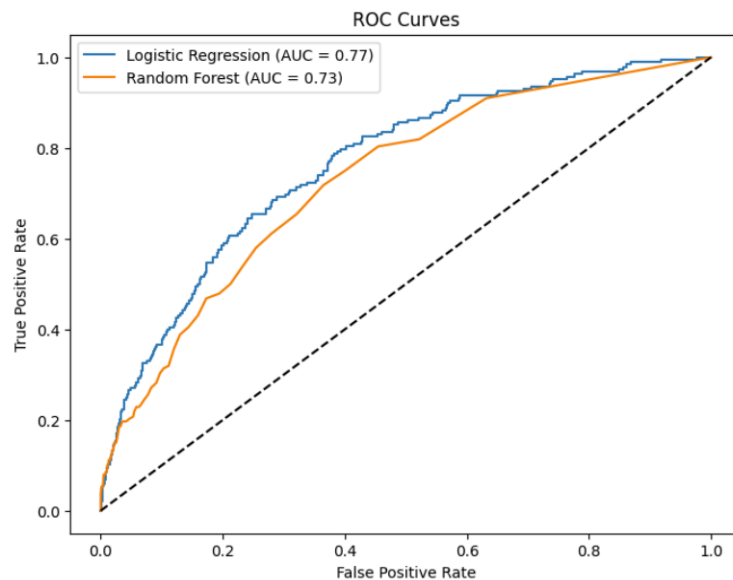
Testing Our Model

We ran the model with the five ratios, and then again with all the values included. We tested three additional financial ratios in this process: liquidity, profitability and leverage. We calculated the VIF to check for variables that may interact with one another, and those with high VIF scores were dropped. The next step was to drop variables that had a p-value of > 0.05 , resulting in just three main factors, X6 (net income), X8 (market value) and X15 (retained earnings). This model was then ran on the test set, and then we created a confusion matrix and ROC curve to visualize the results. These can be seen in our Python code that is linked above, but we resulted with an accuracy of 0.93, and correctly predicted 10 bankruptcies, with a bankruptcy precision value of 0.48 and AUC of 0.77.



Testing Random Forest

Another model we tested as a team was the Random Forest Classifier. This method combines results of different decision trees to make predictions with randomized noise to create a single output. We used 100 estimators and trained the model before running it on the test set. Although it resulted in a similar accuracy, its bankruptcy-yes precision was only 0.41 and AUC was 0.73, leading us to prefer the other model.



Conclusion

While the Altman Z-score provides a solid baseline for bankruptcy prediction, augmenting it with additional financial ratios, particularly those related to liabilities and profitability, enhances its predictive power. However, in the end we dropped them and it came down to three major factors: net income, market value and retained earnings. This was better than the starting model and the Random Forest we tested. This model was more robust, even during periods of financial crisis. Nevertheless, real-world testing with more recent data is needed to validate the model's performance in different market conditions.